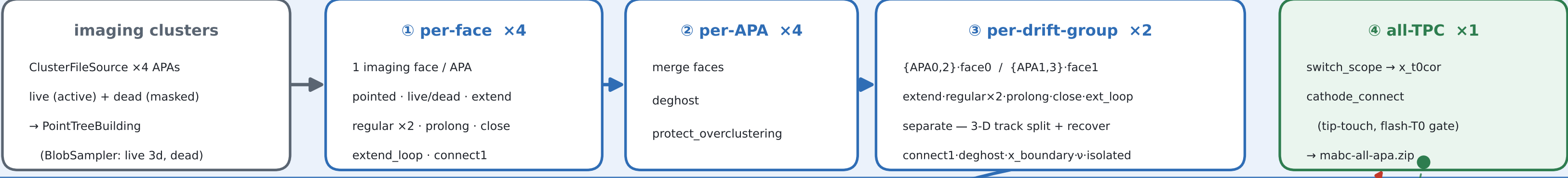


ProtoDUNE Horizontal Drift — Wire-Cell 3-D Clustering + Charge-Light (Q/L) Matching Chain

per-face → per-APA → per-drift-group → all-TPC clustering, with the joint QLMatching node pinning each cluster's drift T0

CHARGE · 3-D CLUSTERING (MultiAlgBlobClustering cascade — live + dead blobs)

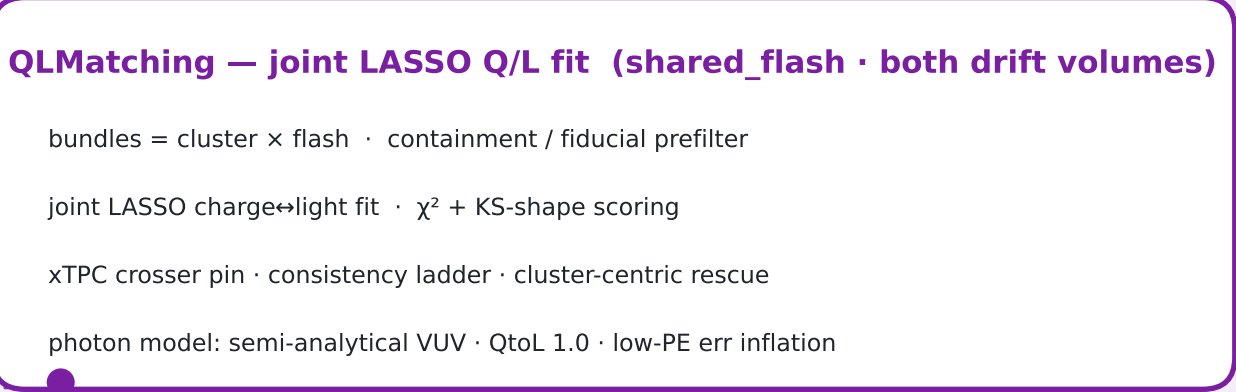


runner operating point `do_qlmatch = on`: ③ → QLMatching → ④ (no-QL: ③ → ④ direct merge)

clusters, both sides

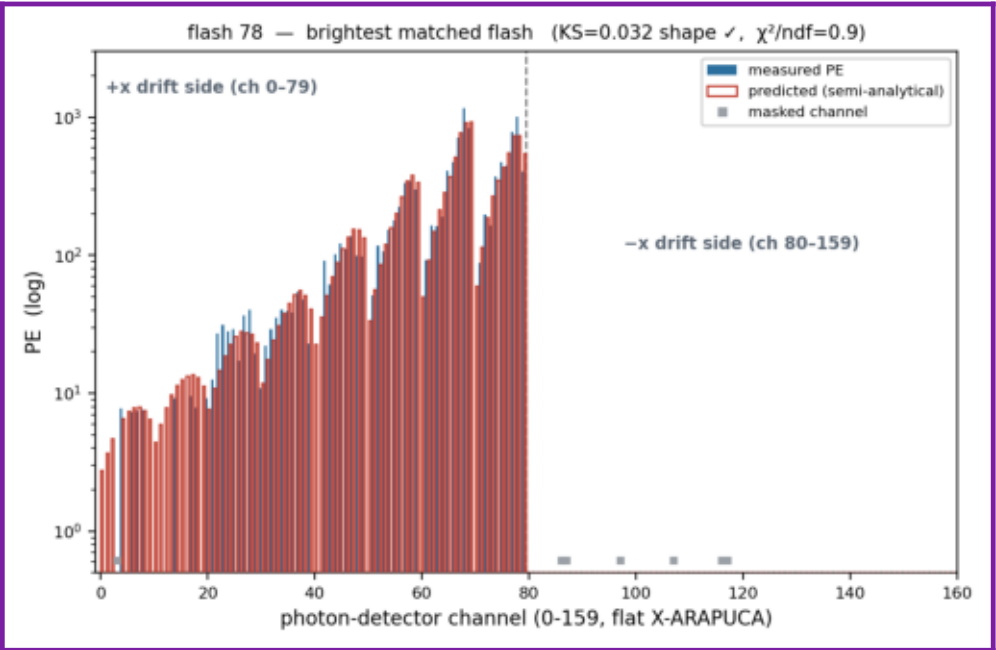
LIGHT RECO (separate wire-cell job)

CHARGE-LIGHT (Q/L) MATCHING

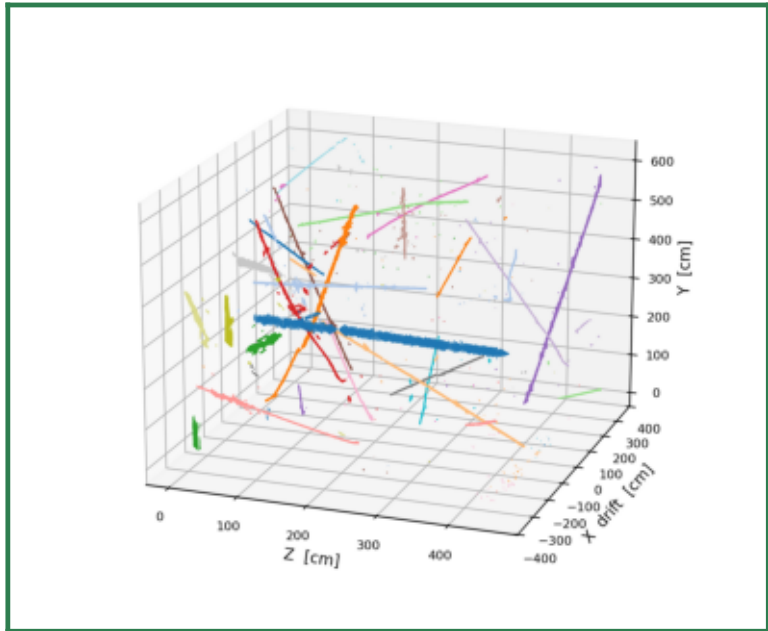


cluster T0 → x_t0cor

- input / imaging
- clustering
- light reco
- Q/L matching
- T0 → x_t0cor



Q/L light pattern — flash 78 (measured vs semi-analytical prediction)



clustering result — clusters coloured (evt 983)